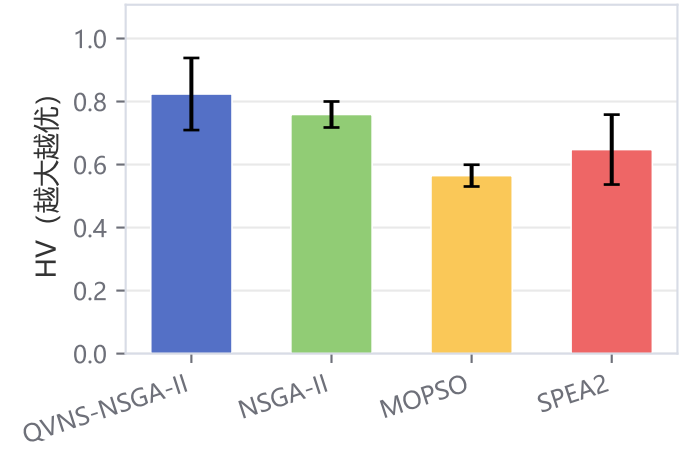
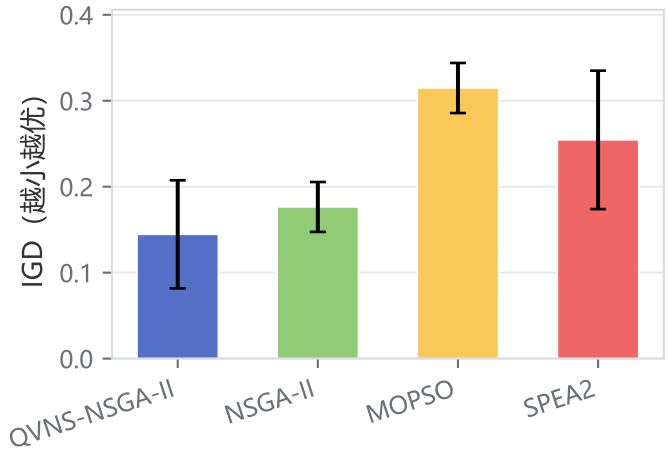
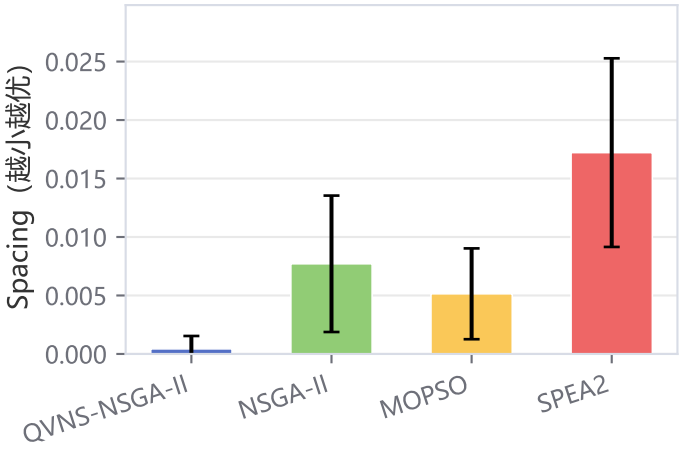
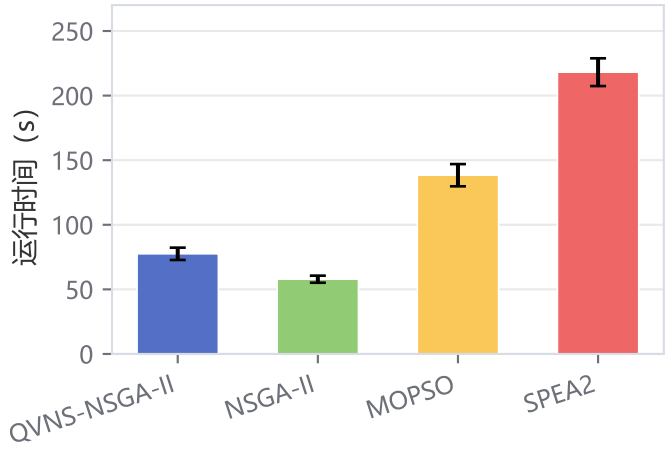

HV（越大越优）
1.0
0.8
0.6
0.4
0.2
0.0
QVNS-NSGA-II
NSGA-II
MOPSO
SPEA2


IGD（越小越优）
0.4
0.3
0.2
0.1
0.0
QVNS-NSGA-II
NSGA-II
MOPSO
SPEA2


Spacing（越小越优）
0.025
0.020
0.015
0.010
0.005
0.000
QVNS-NSGA-II
NSGA-II
MOPSO
SPEA2


运行时间（s）
250
200
150
100
50
0
QVNS-NSGA-II
NSGA-II
MOPSO
SPEA2